

Abstract

- It is a keyword in java
- Abstract Method
 - An 100% incomplete method is an abstract method.
 - in java such methods must be declared as abstract
 - these methods should only have the declaration and not the body.
- Abstract Class
 - the class which is incomplete can be declared as abstract class
 - the class which consists of abstract methods must be declared as abstract class
 - we cannot create instance of an abstract class
 - only references are allowed

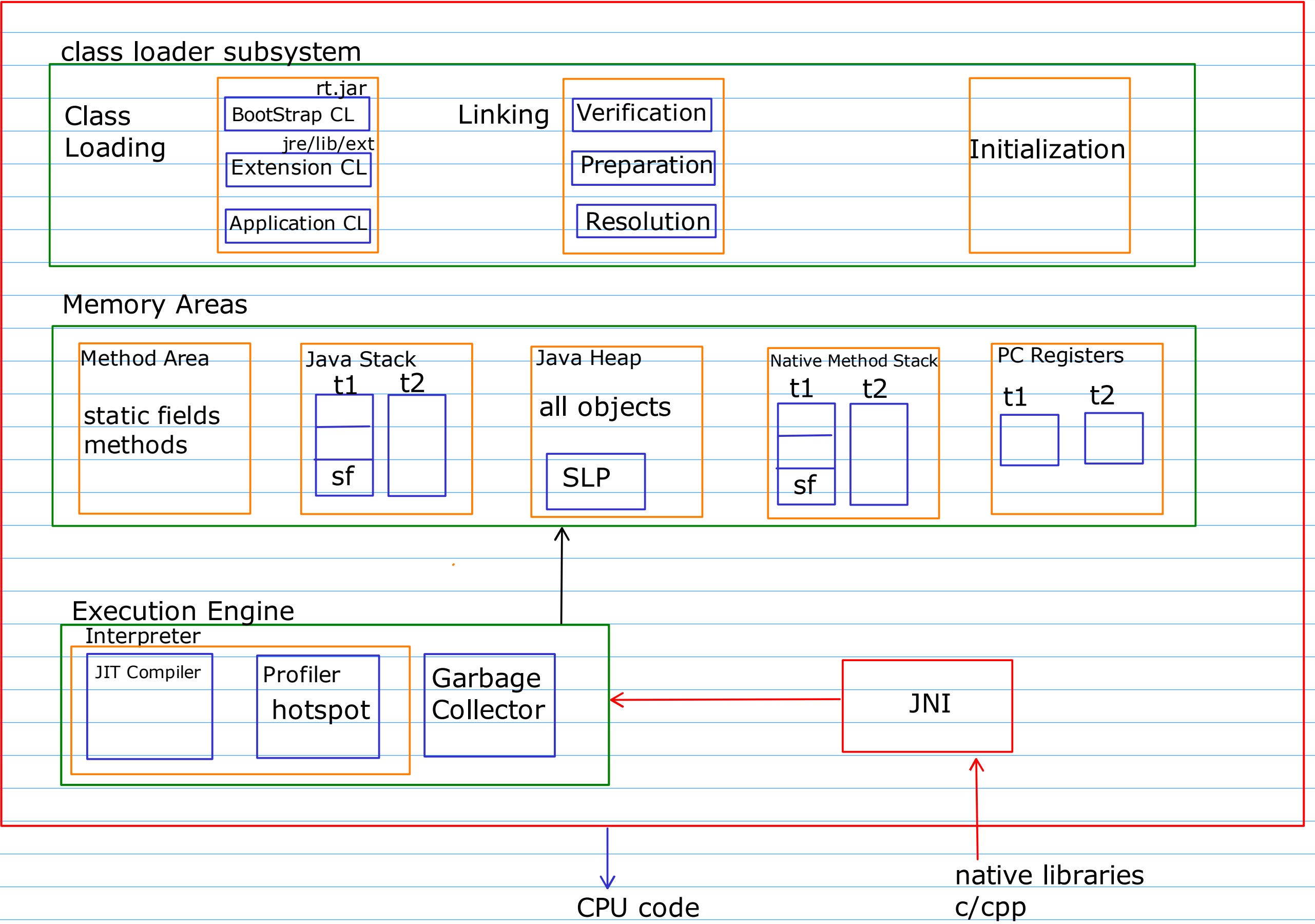
Interface (till Java 7)

- It is designed to group unrelated types together
- It provides the protocols/specification, i.e provides a common behaviour (method design) for the unrelated types
- It prevents the fragile base class problem, by providing the feature of multiple interface inheritance
- inside the interface all the methods that we declare are public abstract
- inside the interface all the fields that we declare are public static final
- inside interface we cannot declare ctor

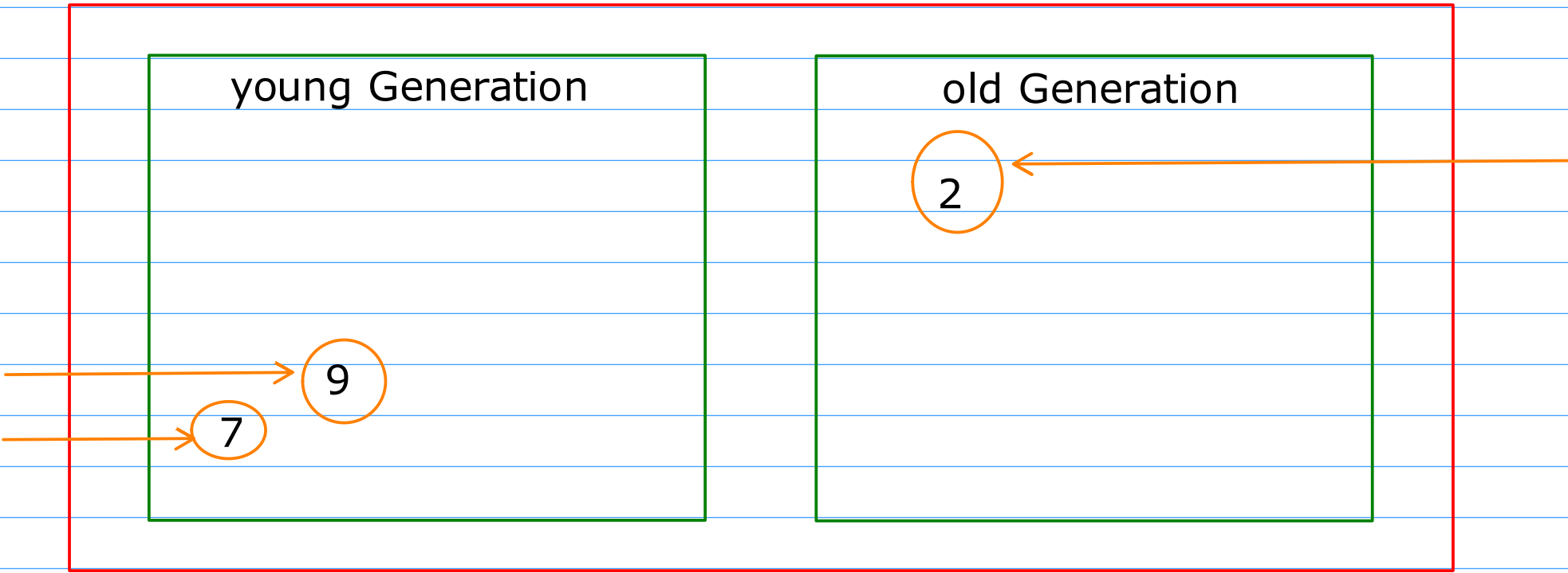
Marker Interface

- An empty interface is called as marker interface
- To group some types together or to provide some permissions
- eg -> Cloneable
 - > Seralizable

Java Virtual Machine(JVM)



Heap



Main Garabage Collection (background)

```
Test t1 = new Test(); // GC
t1 = null;
```

```
Test t1 = new Test(); // GC
t1 = new Test();
```

```
void m1(){
Test t1 = new Test(); // GC    SF
}
```

```
main(){
m1();
}
```

- GC invoked
- Minor GC
 - Major GC

Mark and compact

